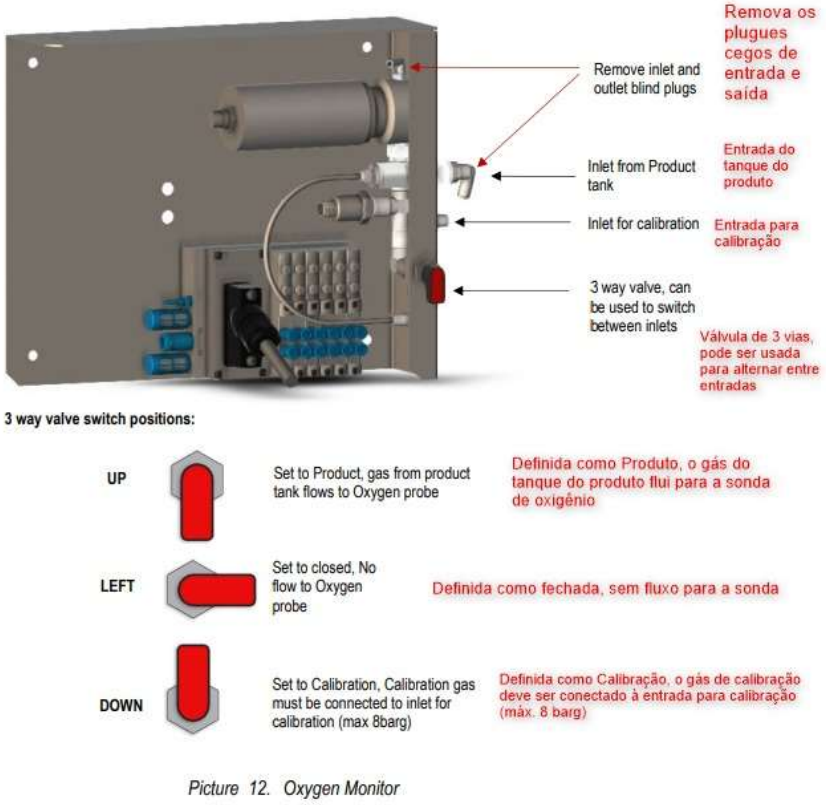
USINAS DE OXIGÊNIO – OXYMAT (O10 a O1500)

<u>Inicialização normal</u>

- Ligue a fonte de alimentação do secador de ar (se presente)
- Deixe a temperatura do secador de ar de refrigeração estabilizar no valor predefinido +3°C
- Ligue a fonte de alimentação do compressor (se presente) ou abra para o suprimento central de ar comprimido de alimentação
- Observe se a pressão do ar de alimentação está acima dos requisitos mínimos de pressão.
- Ligue a fonte de alimentação do gerador de oxigênio PSA
- Inicie o gerador PSA
- Observe se os drenos estão funcionando
- Observe se a pressão de pico do ciclo não excede o valor declarado no Certificado de Revisão de Projeto
- Depois de um tempo, observe que o gerador PSA para automaticamente, quando a pressão do tanque do produto atinge o ponto de ajuste de pressão indicado pelo fabricante no Certificado de Revisão de Projeto
- Observe também que o gerador PSA reinicia automaticamente após cerca de 0,5 queda de pressão, a menos que indicado de outra forma pelo fabricante.

Se o gerador PSA não iniciar ou parar assim, consulte a seção para procedimento e ajuste do ponto de ajuste de pressão



Picture 12. Oxygen Monitor

5 Operation

5.1 Initial start-up procedure

To start the PSA oxygen generator for the first time, follow the steps below:

1. Remove transport brackets at the air dryer (if present).
2. Turn on the power supply to the air dryer (if present).
3. Allow the refrigeration air dryer temperature to stabilize at the pre-set value +3°C
4. Remove transport brackets at the compressor (if present).
5. Turn on the power supply to the compressor (if present).
6. Make sure, that all shut off valves between the plant items are open
7. Let the compressor keep running and observe that the compressor shifts to unloaded operation, when the compressor pressure has reached the pressure stop set point.
8. Carefully and gentle open for feed air supply to the PSA generator.
9. Turn on the electric power supply to the PSA generator.
10. Start PSA generator by pressing START button (IntelliControl) or by pressing manual on/off button (DCP controller only). Refer to control system manual.
11. *Ensure that the drain system works properly* by checking, that air vents from the drain water outlet for about 3 second every 5 minutes.
12. Be aware that air and condensate might be exhausted automatically from compressor and air dryer during start up – this is not a malfunction.
13. Safety valves check according to chapter 6.12
14. Shut off the oxygen application/consumption and *prepare for Leak test*:
    a. Let the plant run in manual mode, until the product tank pressure reaches a minimum of 5 bar.
    b. Stop PSA generator, and the PSA generator will stop after a little while, when the actual operation cycle is completed.
    c. Shut off the feed air supply.
    d. Check if 3 way valve on Oxygen monitor is set to CLOSED position (See picture 12.)
    e. Read and note the pressure P1 in product tank, column 1, column 2, and air tank.
    f. Let the plant stand still (rest) and isolated in pressurized condition for an hour.
    g. After an hour read and note the pressure P2 in product tank, column 1, column 2, column 3 (if available) and feed air tank.
    h. Then determine an eventually pressure drop as the difference between P1 and P2 for each component.
    i. For identification of leaks, use appropriate leaktest fluid on the joints.
    j. *The Leak test is right, if the pressure drops after one hour pressurized isolation is less than 0.1 bar(g). In case of leaks they must not cause more than 0.1 bar(g) pressure drop per hour.*